Subject Description Form

Subject Code	ME5510
Subject Title	Thermal Engineering
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Students should have basic knowledge in Thermofluids
Objectives	To provide students with knowledge of engineering thermodynamics and heat transfer; to enable the students the ability of modeling, analyzing and solving the practical problems in thermal engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. possess state-of-the-art knowledge and skills in the area of thermal science and engineering, be able to apply their knowledge and skills in designing and developing products or engineering systems; b. think critically and holistically in dealing with thermal problems, and generate practical solutions; and c. recognize the need for, and engage in life-long learning.
Subject Synopsis/ Indicative Syllabus	***Engineering Thermodynamics:*** Introduction to thermodynamics and applications; Systems; Properties; Extensive and intensive properties; Measured properties; Energy; Equilibrium; Zeroth law of thermodynamics; State and state principle; Process and cycle; Quasiequilibrium process; Pure substances; Phase-change process; Property diagrams; Critical point and critical properties; Enthalpy; Property tables; Reference state; State equations for vapor phase; Ideal gases; Heat and work; Adiabatic processes; Mechanical forms of work; Polytropic processes; Specific heats; Internal Energy and enthalpy; First law of thermodynamics; Energy balance; Control volumes; Conservation of mass; Conservation of energy; Flow work and total energy of a flowing fluid; Steady-state and unsteady-flow processes; Applications in nozzles, diffusers, turbines, compressors, values, mixing chambers and heat exchangers; Thermal reservoirs; Power cycles; Efficiency; Refrigeration and heat pump cycles; Coefficients of performance; The Kelvin-Planck statement; The Clausius statement; Reversible and irreversible processes; Carnot principle; The Carnot cycle; Thermodynamic temperature scale; Maximum performance measures for cycles; Quality of energy; Clausius inequality; Entropy; Thermal entropy flux and entropy production; Entropy of substances; Entropy changes; Entropy balance for closed systems; Principle of entropy increase; Entropy rate balance for open systems. ***Heat Transfer:*** Heat; Conduction and convection; Fourier law of heat conduction; Thermal conductivity; Heat conduction equation; Newton law of cooling; Convective heat transfer coefficient; Non-dimensional numbers; Double-pipe heat exchangers; Shell-and-tube heat exchangers; Cross-flow heat exchangers; Overall heat-transfer coefficient; Log mean temperature difference; Effectiveness-NTU method; Design and applications.

Teaching/Learning Methodology	1. The teaching and learning methods include lectures/tutorial sessions, homework assignments, test, case study report and examination. 2. The continuous assessment and examination are aimed at providing students with integrated knowledge required for thermal science and engineering. 3. Technical/practical examples and problems are raised and discussed in class/tutorial sessions.

Teaching/Learning Methodology	Intended subject learning outcomes		
	a	b	c
1. Lecture	√	√	√
2. Tutorial	√	√	√
3. Homework assignment	√	√	√
4. Project report and presentation	√	√	√

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
		a	b	c
1. Homework assignment	20%	√	√	√
2. Test	20%	√	√	
3. Projectreport and Presentation	20%	√	√	√
4. Examination	40%	√	√	√
Total	100%			

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

0.40 × End of Subject Examination + 0.60 × Continuous Assessment

The continuous assessment consists of three components: homework assignments, test, and project report & presentation. They are aimed at evaluating the progress of students study, assisting them in self-monitoring of fulfilling the respective subject learning outcomes, and enhancing the integration of the knowledge learnt.

The examination is used to assess the knowledge acquired by the students for understanding and analyzing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.

Student Study Effort Expected		
	Class contact:	
	▪ Lecture	36 Hrs.
	▪ Tutorial/Case study	3 Hrs.
	Other student study effort:	
	▪ Self Study	45 Hrs.
	▪ Project report preparation and presentation	26 Hrs.
	Total student study effort	110 Hrs.

Reading List and References	1. Cengel Y. A. and Boles M. A., *Thermodynamics: An Engineering Approach*, McGraw-Hill, latest edition. 2. Holman J. P., *Heat Transfer*, McGraw-Hill, latest edition. 3. Cengel Y. A., *Heat Transfer: A Practical Approach*, McGraw-Hill, latest edition. 4. Morris W. D., *Heat Transfer and Fluid Flow in Rotating Coolant Channels*, Wiley, latest edition. 5. Han J. C., Datta S., Ekkad S., *Gas Turbine Heat Transfer and Cooling Technology*, CRC Press/Taylor & Francis, latest edition. 6. Yeh L.T. and Chu R. C., Thermal Management of Microelectronic Equipment: Heat Transfer Theory, Analysis Methods, and Design Practices, ASME Press, latest edition. 7. Patankar S. V., *Numerical Heat Transfer and Fluid Flow*, McGraw-Hill, latest edition. 8. Fletcher C. A. J., Computational Techniques for Fluid Dynamics: A Solutions Manual, Springer-Verlag, latest edition.
Last Update	June 2024